L&T Technology Services

ENGINEERING **THE CHANGE**

PID No: 00000

10th Edition TECHgium®

Presentation Round

Purposeful.
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GainShield

Safe AI tuning of PI gains, with a stability certificate and runtime fallback, for bidirectional EV-charger DC links

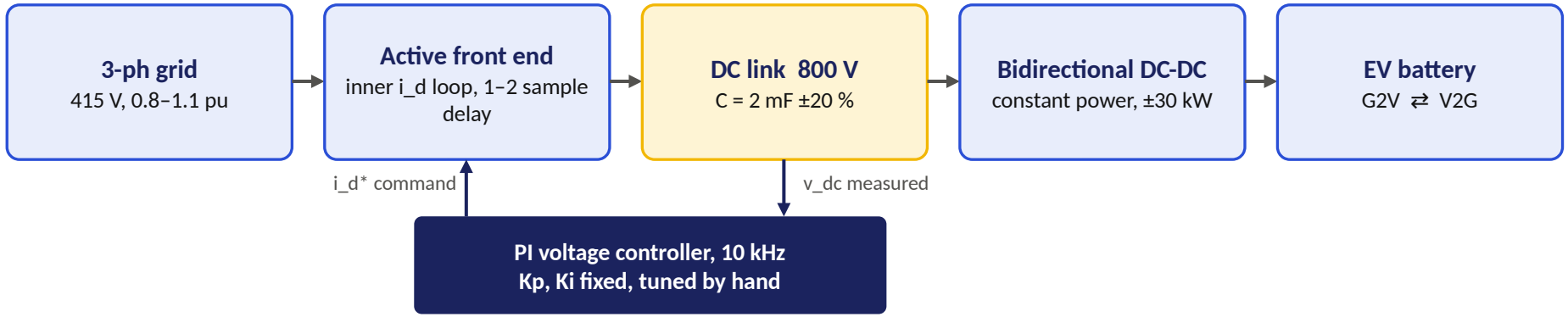
Author 1 Photo	Author 2 Photo	Author 3 Photo	Author 4 Photo
Name 1	Name 2	Name 3	Name 4

College name, City, State

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Challenge Statement

The DC-link PI loop is tuned once by hand, but the plant it controls changes in every charging session



Why it matters

DC-link voltage is the energy buffer between grid and car.

Too slow: voltage dips or swells during power steps and G2V⇄V2G reversal.

Too aggressive: oscillation, trips, stressed semiconductors.

1 The plant moves

C drifts ±20 % with tolerance and ageing; grid sags to 0.8 pu; power reverses from +15 to −15 kW in 15 ms; the DC-DC stage behaves as a constant-power load.

2 Manual PI = one compromise

Tuned for nominal conditions, then detuned to survive the worst case. Our baseline keeps only 42° worst-case phase margin and is not re-tuned when conditions change.

3 Unrestricted AI is unsafe

Bayesian optimisation or RL that tries any gains on a live 800 V link can apply an unstable pair before it learns it is unstable.

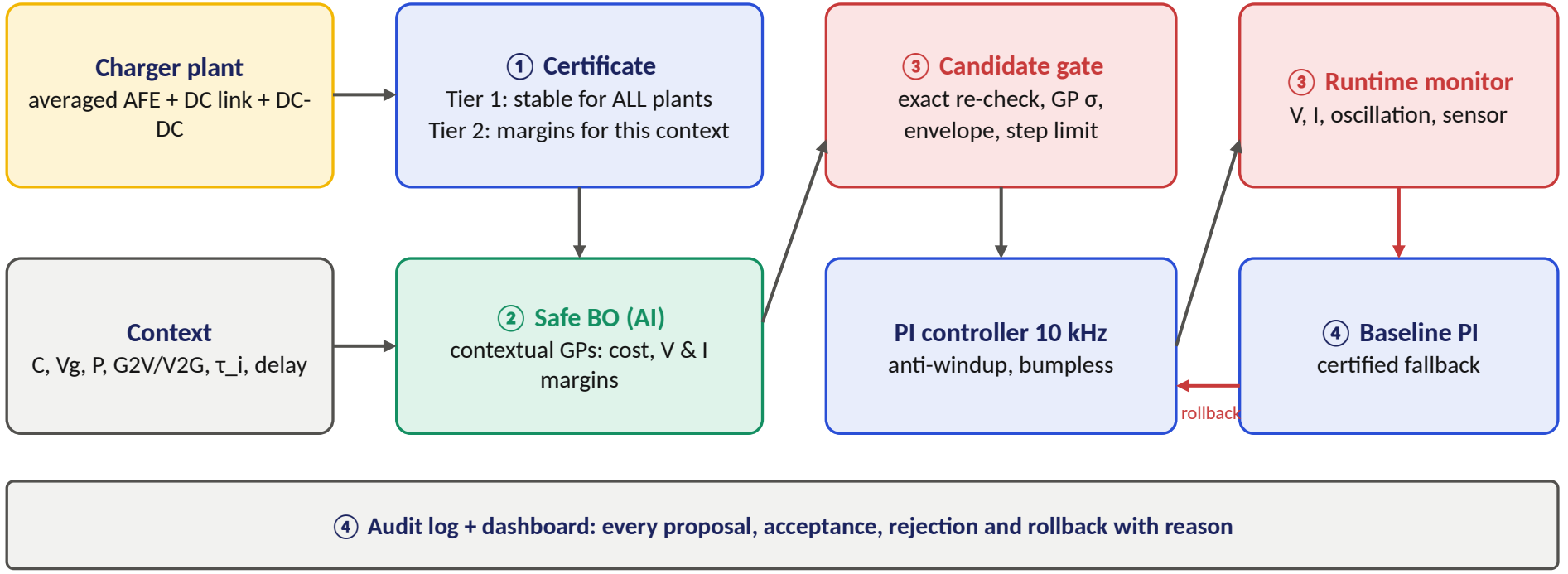
4 No evidence trail

Nobody can show afterwards which gains ran, why, or whether they were proved stable at the time.

Challenge: tune the PI gains online for the current operating point, apply only gains that are proved stable, fall back instantly when something looks wrong, and log every decision.

Concept / Proposed Solution

GainShield: the AI chooses where to explore, while maths and an independent supervisor decide what is allowed



1 Prove

No gain reaches the plant without a stability proof over C ±20 %, 0.8–1.1 pu grid, both power directions.

2 Learn safely

SafeOpt-style BO expands a GP safe set outward from the known-good baseline.

3 Supervise

An independent gate and monitor that do not trust the optimiser.

4 Fall back + log

Bumpless rollback to the certified manual PI, with every decision audited.

Core innovation

A two-tier certificate (a global robust-stability proof plus context-specific margins) is wrapped around contextual safe Bayesian optimisation, so the AI can raise gains when the context allows it, and a sudden grid or capacitance change can never make the active loop unstable.

	Manual PI	Plain BO / RL	GainShield
Adapts to operating point	×	✓	✓
Stability proved before applying	✓ (one point)	×	✓ (whole set)
Runtime fallback + audit	–	×	✓

Pros and Cons of the solution

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Measured strengths, with every limitation stated and a mitigation for each

Strengths (evidence from simulation)

Adaptive tuning

16 % lower mean peak deviation and 15 % lower IAE than the manual PI ($p < 1e-7$).

Stability-aware optimisation

Every applied gain is robustly stable for the whole uncertainty set (proof, not sampling).

Safety supervision

0 of 175 online probe episodes left the ±5 % band; an injected unstable gain was rolled back.

Traceability

834 audit records; an automatic check found 0 unexplained gain changes.

Reproducible testing

Seeded, pre-drawn noise; one command regenerates all results; 36 automated tests.

Limitations → mitigation

Model dependency

The certificate holds only for the linearised model inside the stated set → context-envelope check, monitor, HIL validation.

Computational requirements

Uncached context certificate ≈ 1.6 s in Python → off the 10 kHz loop; precomputed lookup table.

GP uncertainty

A GP is not a guarantee (the RKHS bound is not verified) → GP is one of three layers; σ gate; zero-prior margins.

Stability-model assumptions

Inner-loop lag and delay are gridded, not proved between nodes → margin slack; delay identification in HIL.

Simulation-to-hardware gap

No hardware yet; a stuck sensor is not detected → power-balance observer, hardware OVP/UVP, PIL→HIL roadmap.

Trade-off made explicit: higher gains amplify sensor noise (current-reference ripple +8.5 % on average), and GainShield stays on the baseline when it has no certified, evidence-backed improvement.

Technical Description

Plant model and controller, and why the stability certificate is an exact test

Averaged plant (energy balance, CPL in G2V)

$C \cdot v \cdot dv/dt = P_{afe} - P_{dc} - v^2/R$

$P_{ac} = 1.5 \cdot Vg \cdot i_d$, $P_{afe} = \eta \cdot P_{ac}$ (G2V) or P_{ac}/η (V2G)

$\tau_i \cdot di_d/dt = i_d^*(t - T_d) - i_d$, $\tau_{dc} \cdot dP_{dc}/dt = P_{dem} - P_{dc}$

PI controller (sampled at 10 kHz, ±80 A clamp)

$u[k] = Kp \cdot e[k] + x_I[k]$, $x_I \leftarrow x_I + Ki \cdot Ts \cdot e$ (clamping anti-windup)

Bumpless gain change: $x_I \leftarrow u_{prev} - Kp_{new} \cdot e_{prev}$

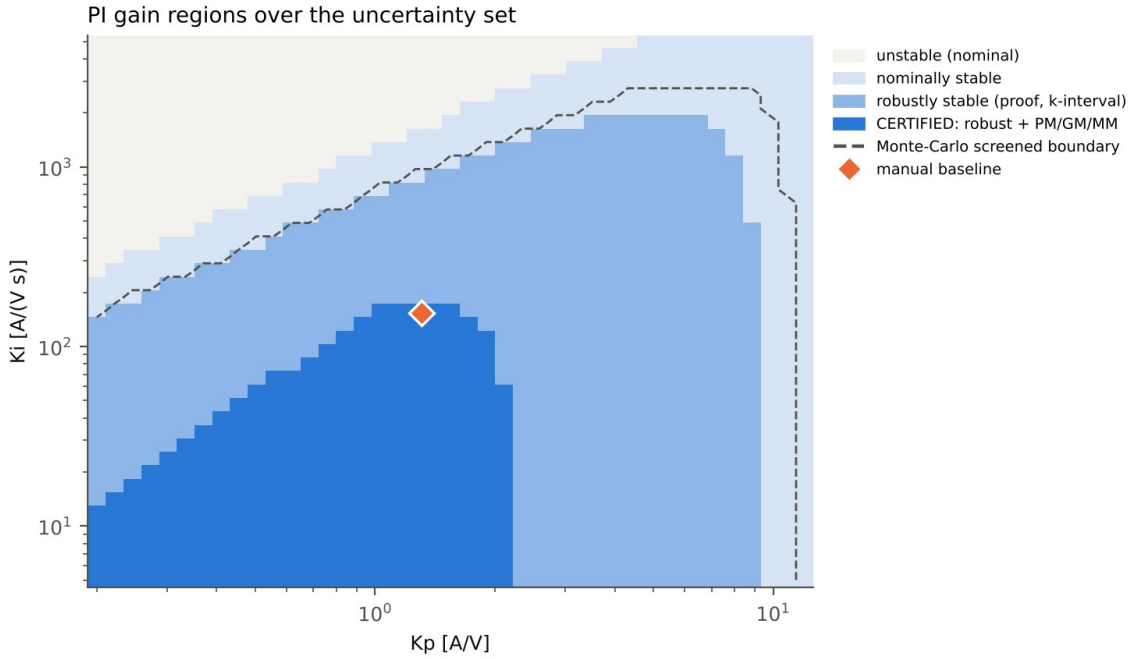
Exact sampled loop (ZOH + fractional delay)

$P(z) = k \cdot \hat{N}(z) / ((z-1)(z-\lambda)z^{(d+1)})$, $k = 1.5 \cdot Vg \cdot \eta / (C \cdot Vref)$

Closed loop: $p(z; k) = D(z) + k \cdot N(z)$ ← affine in k

C ±20 %, grid 0.8–1.1 pu and power direction enter only through $k \in [208, 446]$. Zero-exclusion principle: stable at k_{min} and no unit-circle crossing for any k in the interval ⇒ stable for the whole interval. The crossings are the unit-circle roots of $D \cdot rev(N) - rev(D) \cdot N$.

Checked at 88 inner-lag × delay nodes; margins PM ≥ 40°, GM ≥ 6 dB, $|1+L| \geq 0.5$.



22 %

of the gain box is globally certified

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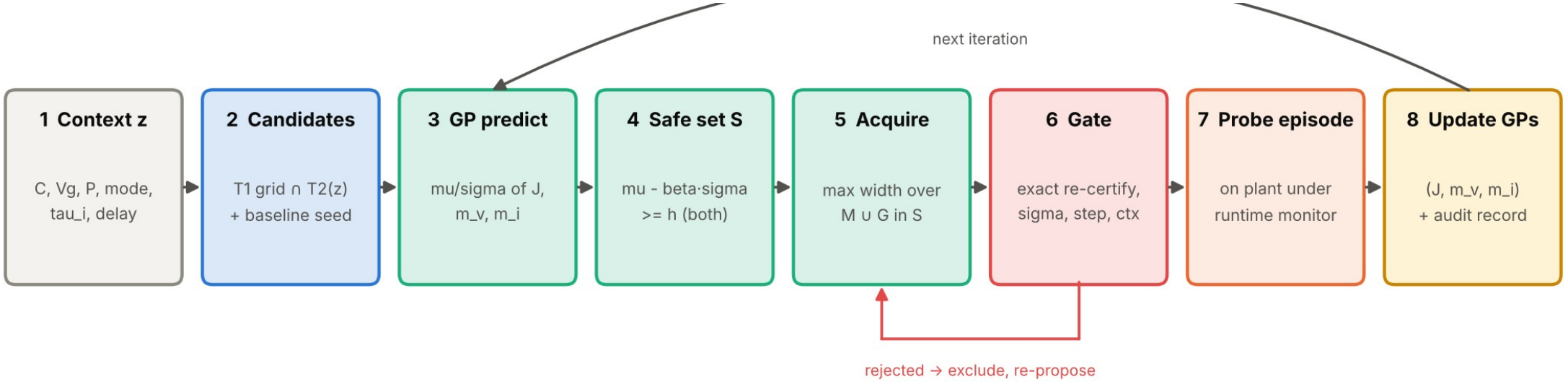
nonlinear instability onsets predicted within ±3 %

42°

manual baseline worst-case phase margin

Technical Description

Safe Bayesian optimisation and an independent runtime supervisor



Contextual GPs (scikit-learn)

Inputs: Kp, Ki (log), C, Vg, P, τ_i, delay.

Outputs: probe cost J, voltage margin m_v, current margin m_i.

Margin GPs: zero prior ⇒ unexplored = unsafe; safe set grows only from the baseline.

One iteration: 60 ms median on a CPU (no GPU needed).

Candidate gate (does not trust the optimiser)

Exact tier-1 + tier-2 re-certification of the gain pair.

GP σ ≤ 0.25 and LCB(margins) ≥ 0.15.

Context inside the validated envelope; gain step ≤ ×3.

Rejected 35 of 210 proposals while tuning, each with a logged reason.

Runtime monitor + fallback (every 100 µs)

|Vdc − Vref| > 4 %, current > 97 % for 2 ms, filtered oscillation, implausible dV.

Alarm ⇒ bumpless rollback to the certified baseline, 100 ms lock-out, then re-decide.

Deployed: gains re-scheduled when Vg, P or the mode changes.

Implementation Plan

Implementation mapped to the six required deliverables: D1 to D6 are built and tested in simulation; HIL is proposed

D1	D2	D3	D4	D5	D6
Plant, constraints, disturbances, manual PI	**Online tuning + stability/safety monitor**	**Fallback, rollback, audit log**	**Simulation dashboard**	**Evaluation vs manual PI**	**Ablation, failure analysis, HIL roadmap**
Averaged RK4 model; 10 seeded disturbance classes; symmetric-optimum baseline	Two-tier certificate; contextual safe BO; candidate gate	Bumpless rollback; lock-out; JSONL audit with trace check	Streamlit: monitoring, AI, safety, experiments, audit, live tuning	11 scenarios × 5 seeds, paired tests, true-plant margins, timing	7 ablation variants; 13 fault cases; HIL plan
Done (sim)	Done (sim)	Done (sim)	Done (sim)	Done (sim)	Done (sim) / HIL proposed

Next phases (proposed, not yet executed)

P1 — Software-in-the-loop

Port the PI, monitor and fallback to C; precomputed tier-2 lookup table

4 wks

P2 — Processor-in-the-loop

TI C2000 F28379D runs the 10 kHz loop; PC runs the optimiser and gate

4 wks

P3 — Controller HIL

Real-time plant simulator; tests H1–H8 (timing, faults, communication loss)

6 wks

P4 — Scaled power bench

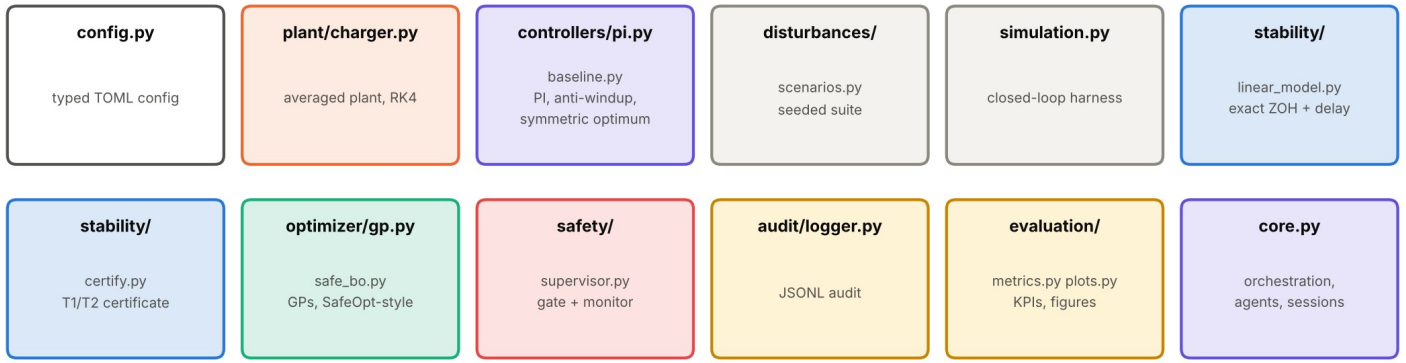
48 V / 1 kW bidirectional converter, low-voltage only

future

Implementation Plan

A modular, tested, one-command-reproducible Python repository

Software modules (package gainshield/)



Outside the package: experiments/ (reproducible scripts), dashboard/ (Streamlit), tests/ (pytest, 36 tests), diagrams/, docs/.

Layer	Technology	Why
Plant + control	NumPy, RK4, 10 µs step	transparent, no licence
Certificate	NumPy polynomial roots	exact zero-exclusion test
Safe BO	scikit-learn GPs + own SafeOpt logic	auditable, CPU-only
Dashboard	Streamlit + Plotly	free hosting tier
Tests	pytest (36 tests)	unit + integration
Reports	pandas, Matplotlib	numbers generated, not typed

python experiments/run_all.py
regenerates the certificate, tuning, evaluation, failure tests, ablation, reports and diagrams in about 15 min on a laptop CPU

GitHub QR code
(to be added when the repository is published)

Repository: gainshield-safe-ev-charger-control

README with results, install and demo guide; documentation of the model, architecture, assumptions, traceability, HIL roadmap, costs and limitations.

No credentials and no machine-specific paths; fixed seeds; pinned versions in requirements-lock.txt.

- **Model-in-the-loop simulation** completed
- **SIL on target code** proposed
- **PIL / controller HIL** proposed
- **Power hardware** future

Validation / Testing / Analysis

Test protocol: identical conditions for both controllers, 11 disturbance scenarios × 5 seeds

#	Scenario	What changes	Class
1	Load increase	battery power 5 → 20 kW	load
2	Load decrease	20 → 5 kW	load
3	G2V → V2G	+15 → −15 kW (slew 2 kW/ms)	reversal
4	V2G → G2V	−15 → +15 kW	reversal
5	Grid sag	1.0 → 0.8 pu at 20 kW, recovery	grid
6	Sensor noise	3× noise + load step	noise
7–8	Capacitance	C −20 % / +20 % + load step	model
9	Inner-loop lag	τ_i = 1.0 ms (2× nominal)	model
10	Control delay	2.0 samples	model
11	Combined	C −20 %, τ_i 1 ms, delay 2, 0.85 pu, 2× noise, reversal	stress

Identical for both controllers: initial steady state, plant parameters, event times, pre-drawn noise sequence, context-estimation error, 0.3–0.4 s duration, Ts = 100 µs, constraints. Metrics are computed on the true plant states.

Metrics (required by the problem statement)

Settling time (±0.5 % band), peak deviation, recovery overshoot, IAE
Voltage violations (±5 %), trips (±10 %), current-limit violations
Stability margin of every gain set used, on the true plant
Computation time per step, per decision, per BO iteration
Robustness: C ±20 %, lag, delay, noise ×2/×3; power reversal

-15.6 %
mean peak deviation

-14.9 %
mean IAE

-7.9 %
mean settling time

0 ms
outside ±5 % (both)

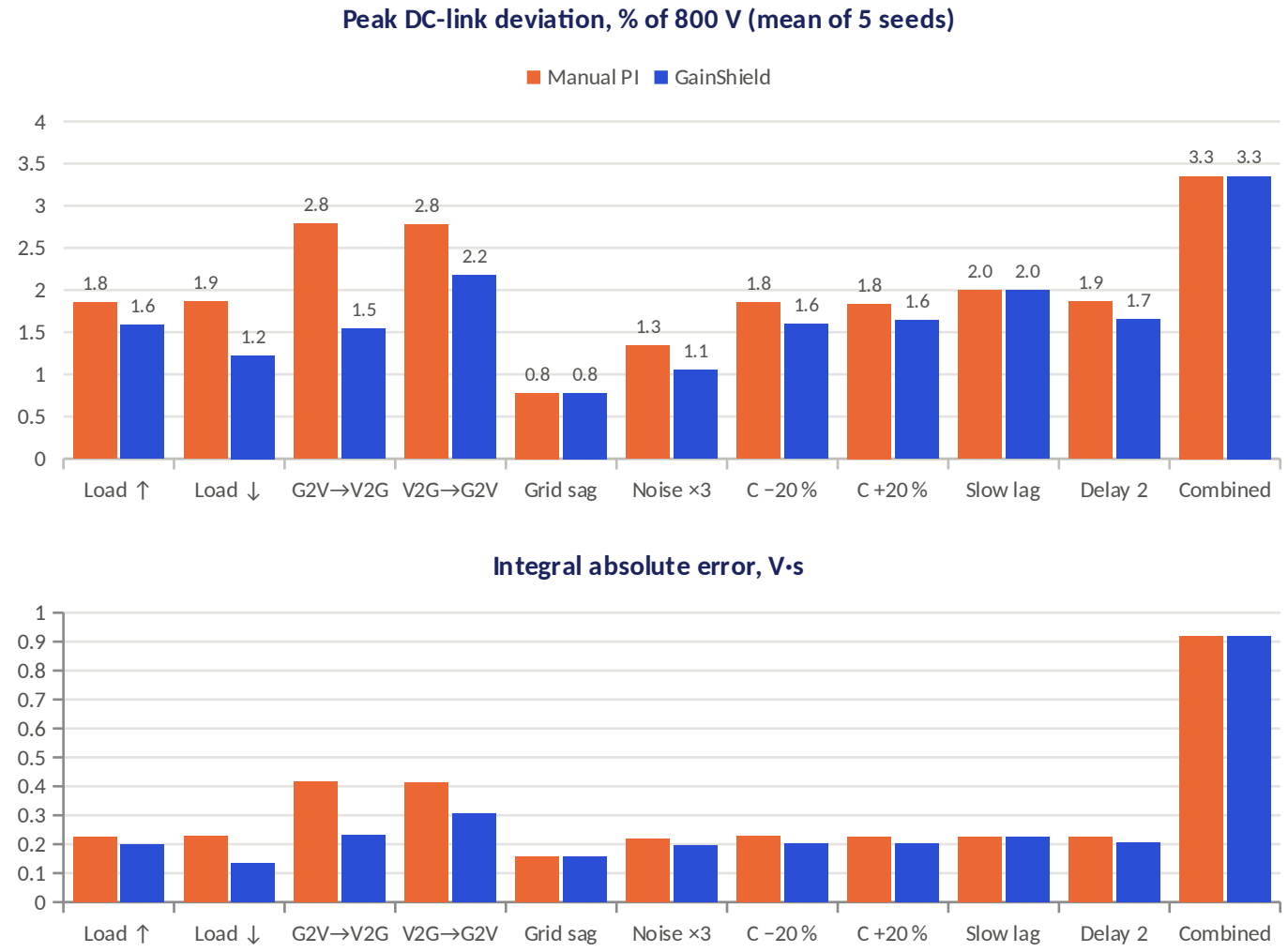
0
trips or current violations

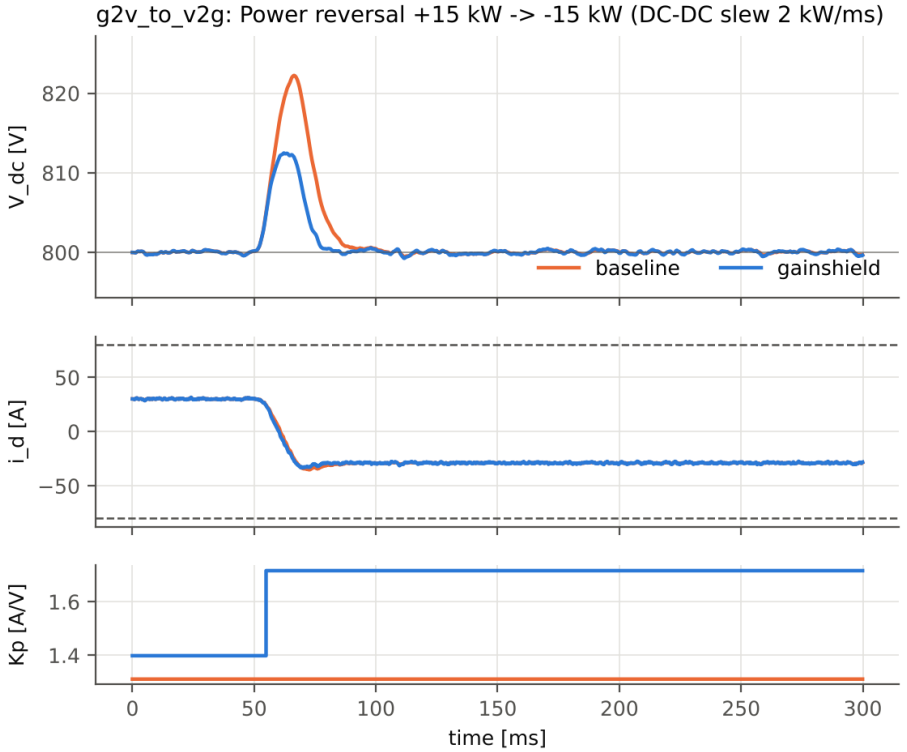
p = 4e-08
Wilcoxon, 55 pairs

GainShield vs manual PI, 11 scenarios × 5 seeds. Source: results/evaluation_summary.json (experiments/run_evaluation.py)

Validation / Testing / Analysis

Results: lower voltage deviation in 8 of 11 scenarios; never worse, because it stays on the baseline when unsure





G2V → V2G reversal: peak 2.79 % → 1.54 %, IAE -44 %. Kp is re-scheduled when the power reverses.

Price paid: current-reference ripple +8.5 %; true-plant PM ≥ 46° (baseline also 46° worst).

Held at baseline: grid sag at 20 kW, slow inner loop and the combined stress (no certified, evidence-backed improvement).

Source: results/evaluation_summary.json, results/figures/eval_g2v_to_v2g.png

Validation / Testing / Analysis

Ablation and fault injection: each safety layer earns its place, and one gap is still open

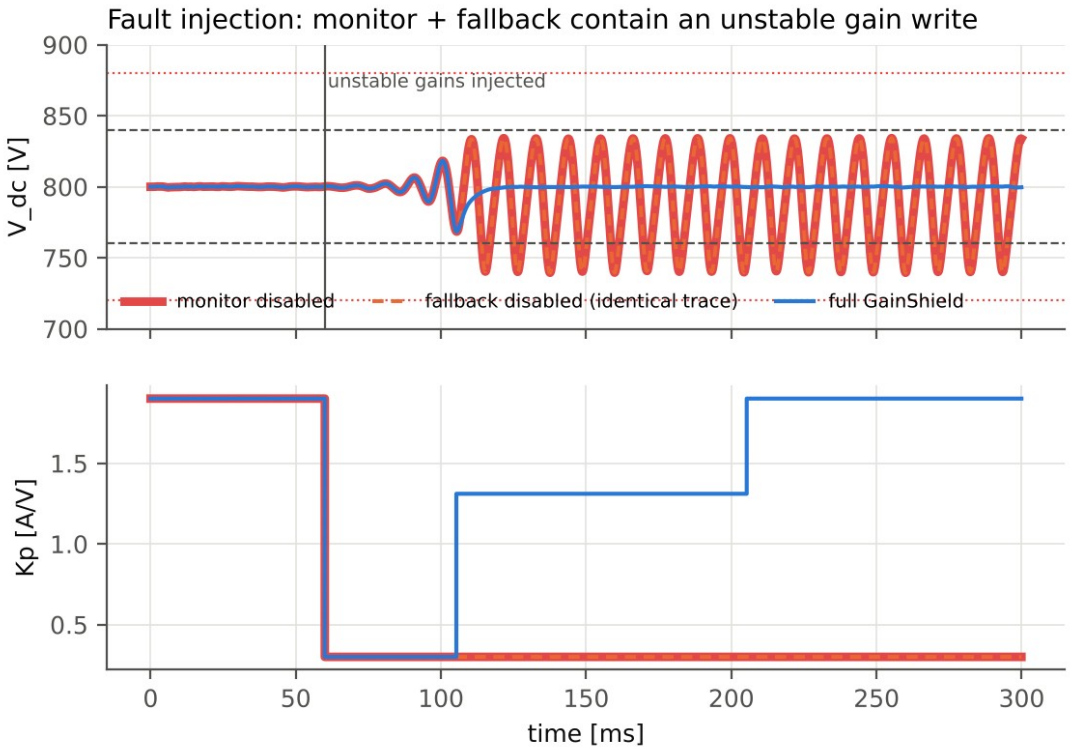
Variant	Unsafe tuning episodes	Deployed peak dev. vs PI	Unstable gains injected: peak / outside ±5 %
A full	0	0.82×	3.9 % / 0 ms
B −cert.	0	0.84×	4.7 % / 0 ms
C −σ	0	0.64×	4.4 % / 0 ms
D −monitor	0	0.82×	7.5 % / 48 ms
E −context	0	0.89×	4.7 % / 0 ms
F −fallback	0	0.82×	7.5 % / 48 ms

D and F: without the monitor or the fallback, an unstable gain write turns into a sustained oscillation; the full system rolls it back.

C: trusting GP means gives faster but unverified gains (only 15 measured episodes); the σ check prevents this.

B: no unsafe episode within this budget, so the certificate's value is a guarantee, not an average.

E: without context the improvement shrinks (0.89× vs 0.82×).



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unsafe proposals rejected by the gate, each with a logged reason (outside the region, σ, LCB, envelope, step)

Open gap

A stuck voltage sensor is not detected (the baseline is equally blind). Mitigation: power-balance observer and hardware OVP/UVP.

Source: results/ablation.csv, results/failure_injection.csv, docs/FAILURE_ANALYSIS.md. Isolated simulation only.

Cost Estimate

The prototype cost ₹0 in cash; the next validation step costs about ₹20,000; HIL through lab access first

Tier	Item	Basis / source (12 Nov 2026, 1 USD ≈ ₹94)	INR
A Actual	Software stack	open-source Python (NumPy, SciPy, scikit-learn, Streamlit)	₹0
A Actual	Computing / cloud / storage	2-core CPU, no GPU; GitHub free tier	₹0
A Actual	Dashboard hosting	Streamlit Community Cloud free tier (public app)	₹0
A Actual	Development effort (in kind)	480 h at an assumed ₹300/h, not a cash cost	≈ ₹1.44 lakh
B Estimate	2 × TI LAUNCHXL-F28379D	$44.16 (Arrow) to ₹5,712 (Indian distributor)	₹8,300–11,400
B Estimate	Signal conditioning, isolation, cables	engineering estimate	₹6,000–10,000
C Optional	Real-time HIL via institute lab	usage charges, if any	₹0–50,000
C Optional	HIL teaching kit (Typhoon HIL101)	US MSRP $9,900; APAC price differs; excl. duties/GST	≈ ₹9.3 lakh
C Optional	Research-grade HIL (HIL404 class)	published 2025 commercial quote, $44k incl. licences	≈ ₹41 lakh
C Optional	Scaled 48 V / 1 kW bidirectional bench	engineering estimate, not quoted	₹1.5–3 lakh

₹0 cash spent on the delivered prototype

≈ ₹20k next phase: PIL validation

No GPU full pipeline ≈ 15 min on a laptop CPU

Sources: Arrow, Typhoon HIL product page, Streamlit docs, currencybeacon.com. Budgetary only: no vendor quotation was requested and no partnership exists. Details: docs/COST_ESTIMATE.md

Assumptions

The problem statement gives no numbers, so every value below is a stated engineering assumption

Group	Assumption (simulation, not a hardware rating)
Plant	30 kW module, 800 V DC link, C = 2 mF ±20 %, 415 V grid 0.8–1.1 pu
Plant	AFE η = 98 % (direction-dependent); inner loop τ_i 0.3–1.0 ms
Plant	DC-DC power loop τ = 2 ms, slew 2 kW/ms, ±30 kW (constant-power load)
Switching	f_sw = f_s = 10 kHz; delay 1.0–2.0 samples (nominal 1.5)
Electrical limits	±5 % operating band, ±10 % trip, 4 % supervisor abort; i_d ≤ 80 A
Sensors	noise 1 V and 0.5 A (1σ); scaled ×2 and ×3 in tests

Group	Assumption
Simplifications	averaged model: no switching ripple, AC filter/PLL, unbalance, battery or thermal dynamics
Stability	linearised and unsaturated; k-interval proved; τ_i × delay gridded (88 nodes); worst-case PM ≥ 40°, GM ≥ 6 dB, MM ≥ 0.5
Context	C, τ_i and delay identified within ±5 %, ±20 % and ±0.2 samples (estimator not implemented)
Optimiser	β = 2, h = 0.15, σ ≤ 0.25, step ≤ ×3; fixed margin-GP hyper-parameters; RKHS bound not verified
Probe	+10 V reference step and ±6 kW power step acceptable during a session
Validation	simulation only; no SIL, HIL or hardware tests performed

All values are in config/default.toml (IDs A-1 to A-15, rationale in docs/ASSUMPTIONS.md); changing them and re-running regenerates every result.

References

References: each one supports a specific design decision

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[15] Typhoon HIL product pages (HIL101, TyphoonSim); Arrow (LAUNCHXL-F28379D); Streamlit Community Cloud docs; accessed 12 Nov 2026.

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